Annex II. Debt Sustainability Analysis

Risk of external debt distress:	Moderate
Augmented by significant risks stemming from domestic public and/or private external debt?	No

The debt sustainability analysis (DSA) shows improved prospects relative to the previous review due to stronger macroeconomic outlook.[1] However, staff still assess the risk of debt distress as moderate.[2] Continued strengthening of institutions underpinning sound macroeconomic policies would be required for a future upgrade.

A. Background

1. **Honduras' public debt increased slightly in 2016**. After satisfactory implementation of the IMF-supported program, nonfinancial public sector (NFPS) gross public debt stood at 42½ percent of GDP in 2016, an increase of about 1 percentage point *vis-à-vis* 2015. In 2016, the fiscal accounts posted a record low deficit of ½ percent of GDP. The NFPS public debt stock presented in this annex updates the stock of unpaid claims identified at the beginning of the program with results of the audit. It also includes the outstanding liabilities from the security tax trust fund around ½ percent of GDP.[3] Additionally, at the onset of the Fund-supported program, the authorities adopted the IPSAS-32 standard for reporting PPP operations in the fiscal accounts, the net stock of liabilities stemming from PPPs is also included into the gross public debt stock (½ percent of GDP).[4] In January 2017 the authorities issued US$700 million in international markets. Because of strong confidence in Honduras' economic outlook supported by the program implementation, the demand for the bonds was around US$5 billion, well above the amount placed. The yield also reached an historical minimum of 6¼ percent (see also EMBI spread chart below).

2. **Public debt is mostly with foreign creditors.** The fraction of public debt that is external rose from 67 percent to about 73 percent after the January 2017 bond issuance. The main external lenders to Honduras are the Inter-American Development Bank (IADB), the Central American Bank for Economic Integration (BCIE), and the World Bank. Debt to these institutions carries long maturities but only the IADB continues to provide loans on concessional terms (i.e., with a grant element of at least 35 percent). Domestic public debt is mainly to commercial banks, has a shorter—though rising—maturity (about 3 years), and carries a higher real interest rate.

[1] See *Annex I* of *Honduras : 2016 Article IV Consultation, Third and Fourth Reviews under the Stand-By Arrangement and the Arrangement under the Standby Credit Facility-Press Release and Staff Report* (IMF Country Report No. 16/362).

[2] Honduras three-year average CPIA score is 3.41 and its policy performance category is medium.

[3] Debt from the security tax trust fund is not recorded as public debt by the authorities because of legal considerations, however, as the fund is financed by an earmarked FTT it should be classified as a general government entity according to the GFSM.

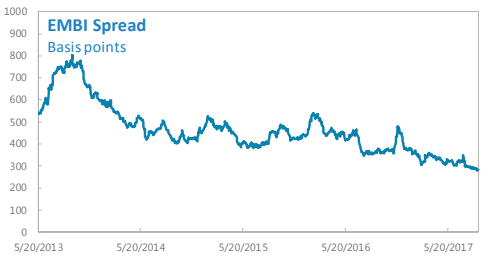
[4] See International Financial Reporting Standards (2013) for details on the IPSAS-32 accounting standard.

B. Underlying Assumptions

3. **Economic growth.** The baseline scenario assumes that growth in Honduras will reach 4 percent in 2017 driven by private consumption, exports and public investment, and moderate to about 3.8 percent in the long-run—in line with closing the output gap in the medium term. This reflects a slightly stronger outlook compared to the previous DSA for 2017, where growth was expected to reach 3.7 percent, on account of stronger-than-expected remittances growth and non-*maquila* exports that are benefiting from better terms of trade and a recovery in coffee production that had been suffering from a coffee rust outbreak. Risks to the macroeconomic outlook are tilted to the downside. Tighter U.S. immigration policy, including deportations, may reduce remittances, increase local labor market pressures and reduce output growth.

4. **Fiscal policy.** As in the previous DSA and consistent with the fiscal responsibility law (FRL) we expect that fiscal projections continue to be anchored by the FRL's rules. The law states a NFPS deficit ceiling of 1 percent of GDP from 2019 onwards and a limit in the nominal increase in current spending (around 7 percent). Our baseline scenario factors in a cumulative fiscal adjustment (measured using the primary balance) of about 7 percent of GDP during the program period (2014–17), with the primary balance becoming consistently positive starting in 2018. The fiscal consolidation included a combination of higher tax revenues (about 3½ percent of GDP) and lower expenditures (about 5½ percent of GDP).[5] On the revenue side, the main elements were an increase in tax rates and the adoption of anti-evasion measures geared to strengthen revenue administration; on the spending side, key elements were the rightsizing in public employment and reshaping the role of the state in strategic sectors such as energy and telecommunications. The fiscal deficit is projected to be within the law's parameters and the authorities medium term fiscal framework (see ¶10).

5. **Fiscal institutions.** Consolidating the role of the FRL as the cornerstone of macroeconomic policies is the critical challenge for the next years. On this front, the experience in Latin America with the adoption of a FRL is mixed. In some countries, a FRL failed to institutionalize fiscal prudence and in others it served to catalyze fiscal reforms and became the backbone of sound policy making. The implementation of the FRL in Honduras will require developing institutions, strengthening commitment, and fostering transparency and accountability across all levels of government. This process would need time to mature and therefore should be considered a medium-term target.



Source: Bloomberg.

6. **Public sector financing.** Financing of the public sector is assumed to come mainly from external sources. The financing mix is expected to change toward less concessional sources compared to recent years (Table A3), including with multilateral institutions and external capital markets, and in line with Honduras increasing ability to tap international markets.

[5] Despite the higher tax collection, the contribution of total revenues to the adjustment was lower (1½ percent of GDP) because of the decline in other revenues particularly ENEE's tariffs which are linked to oil prices.

7. **External sector.** The external current account deficit is projected to decline from 5½ percent of GDP in 2015 to about 4 percent by 2022, and stabilize slightly at around 3½ percent of GDP over the longer term. The external position is expected to improve over the medium to long-term owing to supportive macroeconomic policies and better terms of trade. The improvement in the current account reflects favorable external conditions (including higher coffee and maquila exports), and a stronger macroeconomic policy mix (fiscal consolidation based on reaching the targets stated in the FRL). As in the previous DSA, the current account deficit continues to be financed through foreign direct investment and, to a lesser extent, public sector borrowing, with very small private sector flows. This allows international reserves to increase to about 5 months of non-*maquila* imports by 2018.

C. External DSA

8. **External debt is expected to start decreasing around 2020** (Table AII.1). External debt ratios are projected to peak at about 39 percent of GDP in 2020, from about 36 percent of GDP in 2016. The share of external debt in total public debt rises slowly to about 78 percent by 2020. The increase takes place as total debt decreases substantially due to fiscal consolidation. Private external debt is around 5 percent of GDP.

Selected Macroeconomic Indicators, Current vs. Previous DSS

	2015	2016	2017	2018	Long term 1/
Real GDP growth (percent)					
Current DSA	3.6	3.6	4.0	3.6	3.8
Previous DSA	3.6	3.6	3.7	3.8	3.8
GDP deflator growth (percent)					
Current DSA	6.4	3.7	5.9	3.8	4.4
Previous DSA	6.3	3.0	3.1	4.3	4.5
Primary balance (% of GDP)					
Current DSA	-0.3	-0.3	-0.1	-0.1	0.6
Previous DSA	0.1	0.0	0.1	0.4	-0.1
Current account balance (% of GDP)					
Current DSA	-5.5	-3.8	-4.1	-4.3	-3.7
Previous DSA	-6.3	-5.9	-5.7	-5.7	-4.6
Net FDI (% of GDP)					
Current DSA	5.4	3.7	4.9	4.8	5.2
Previous DSA	5.4	5.7	6.1	6.0	6.4

Source: IMF staff estimates.

1/ Defined as the last 15 years of the projection period. For the current DSA, the long term covers the period 2023-37, whereas for the previous DSA it covered 2022-36.

9. **External debt ratios remain within indicative thresholds in most scenarios.**[6] The ratios for the PV of public debt and public debt service remain below their indicative thresholds in most scenarios throughout the projection period. External public debt remains initially stable and starts to decline earlier than in the previous DSA due to stronger than expected fiscal consolidation. Under most extreme shocks (one-time 30 percent depreciation), the debt service to revenue ratio threshold is marginally exceeded in 2020 and 2027, reflecting the amortizations of global bonds with balloon payments. The ratio of debt service to exports and remittances remain well below its threshold under all scenarios.

10. **Most indicators remain within indicative thresholds in the baseline scenario under the probability approach**. Figure AII.2 reports the results from the probability approach. Most ratios remain within indicative thresholds under the baseline scenario. Under the most extreme shock (an export shock), the threshold for the PV of debt to GDP ratio is breached initially and only returns to levels below the threshold in the late 2020s. The ratio of debt service to revenue is breached only in 2020 and 2027 under the most extreme scenario (a one-time 30 percent depreciation), reflecting the bullet payments of recent bond issuances. Going forward, some of the policy reforms that the authorities have embarked upon would increase the overall flexibility and dynamism of the economy to cushion some of these shocks. The ongoing scaling up of public sector investment in high-priority infrastructure would increase overall economy-wide productivity growth and lead to the development of other sectors, thus helping to make the economy more diversified and resilient to export shocks, such as the one mentioned above.

Baseline vs. Customized Scenario 1/

	2017	2018	2019	Long term 2/
Real GDP growth (percent)				
Baseline	4.0	3.6	3.8	3.8
Customized	2.6	2.5	2.8	3.5
Average interest rate on new external debt				
Baseline	4.3	4.3	4.3	4.3
Customized	5.3	5.3	5.3	5.3
Fiscal balance				
Baseline	-1.0	-1.0	-1.0	-0.6
Customized	-1.0	-3.0	-2.7	-1.0

Source: IMF staff estimates.

1/ In the customized scenario, (i) real GDP growth is 2.6 percent in 2017 and 2.5 in 2018, and rises on average 0.25 percentage points per year to reach 3.5 percent in 2021, staying at this level for the remainder of the DSA period; (ii) financing conditions from external capital markets deteriorate permanently, raising the average interest rate on new external debt by 100 basis points; and (iii) the overall deficit of the CPS increases by 2 percent of GDP in 2018 compared to the baseline and then fades out partially at a constant rate of 15 percent during the DSA period.

2/ Defined as the last 15 years of the projection period. For the current DSA, 2023-37.

11. **External debt indicators appear resilient to a customized scenario combining negative real, fiscal and financing shocks.** The customized scenario aims at capturing possible

[6] Honduras has large remittances inflows — about 18 percent of GDP. Therefore, the indicative external DSA thresholds reported are remittances-adjusted as per the DSA template.

downside risks and key vulnerabilities in Honduras. It includes negative shocks to GDP growth, which could arise from weaker external conditions; tighter external financing conditions that could be associated with negative developments in international financial markets; a weaker fiscal position from possible slippages in the implementation of the fiscal responsibility law.[7] In this scenario, external public debt slightly rises initially but remains below the indicative threshold and starts to decline in the early-2020s. The ratio of debt service to exports and remittances, and to revenue also remain below their indicative thresholds.

D. Public DSA

12. **Public debt ratios are expected to peak in 2019 and then start to decline subsequently** (Table AII.3). Public debt is projected to peak at about 45 percent of GDP in 2019 (up from 43½ percent of GDP in 2016) and start falling slowly as fiscal consolidation proceeds and interest payments decline reaching 33.3 percent of GDP by 2027. In present value terms, the public-debt ratio is expected to peak at around 41 percent of GDP in 2019 and fall to less than 30 percent of GDP in the late 2020s. The public debt dynamics remains somewhat vulnerable to both policy-related and exogenous shocks, especially to those related to economic growth and fiscal policy (Table AII.4). Only the scenario based on historical variables breaches the PV of debt-to-GDP ratio. This is because Honduras ran large fiscal deficits before the Fund-supported program. Since the beginning of the problem, the authorities have shown their commitment to fiscal prudence with a reduction of 7¼ points of GDP in the fiscal deficit in 3 years and their intention to pursue within this line has been clearly stated in the issuance of the fiscal responsibility law. Because of this, staff do not believe that the scenario based on historical behavior of fiscal variables is representative of the future evolution of the Honduran debt dynamics.

13. **The exposure to contingent liabilities seems to be limited**. Several measures recently taken by the authorities limit the exposure of public debt to contingent liabilities. For instance, by doubling the contribution rate for the social security institute, the authorities' have improved its actuarial position; the PPP framework law was amended to limit the provision of government guarantees for PPP operations; and by upgrading PPPs accounting rules to international standards the impact of firm PPP's operations is included into the regular spending or financing. The recent bond issuance is expected to be used to clear all ENEE's firm liabilities. This leaves only one contingent obligation for ENEE due to pre-program claims to energy providers which are still under dispute (estimated at around ¾ percent of GDP).

[7] Specifically, in the customized scenario (i) real GDP growth is 2.6 percent in 2017 and 2.5 in 2018, and rises on average 0.25 percentage points per year to reach 3.5 percent in 2022, staying at this level for the remainder of the DSA period; (ii) financing conditions from external capital markets deteriorate permanently, raising the average interest rate on new external debt by 100 basis points; and (iii) the overall deficit of the CPS increases by 2 percent of GDP in 2018 compared to the baseline and then fades out partially at a constant rate of 15 percent during the DSA period.

E. Conclusion

14. **Despite marked improvements on public and external debt indicators, including a satisfactory Fund-supported program implementation, the risk of external debt distress remains moderate.** Stronger than expected fiscal consolidation, and a reduced current account deficit strengthen the resilience of Honduras external position. The FRL passed in 2016 is a great achievement and sets the roadmap for continued structural fiscal reforms over the coming years. On the monetary front, the reform of the monetary policy framework has shown some progress but much more needs to be done. Proper implementation of such reforms over the next years are critical to generate a solid track record of commitment to sound macroeconomic policies, therefore, inclining future risk assessments towards a low distress rating.

Table AII.1. Honduras: External Debt Sustainability Framework, Baseline Scenario, 2014-37 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2014	2015	2016			2017	2018	2019	2020	2021	2022	2017-2022 Average	2027	2037	2023-2037 Average
External debt (nominal) 1/	**37.2**	**36.5**	**35.9**			**38.2**	**37.9**	**38.4**	**38.6**	**37.9**	**37.2**		**31.0**	**19.2**	
of which: public and publicly guaranteed (PPG)	29.0	29.1	29.4			32.2	32.3	33.1	33.3	32.7	32.0		25.9	14.3	
Change in external debt	0.8	-0.7	-0.6			2.3	-0.3	0.5	0.2	-0.7	-0.7		-1.3	-1.0	
Identified net debt-creating flows	-2.0	-1.7	-1.0			-2.1	-1.8	-2.3	-2.4	-2.5	-2.4		-2.5	-2.5	
Non-interest current account deficit	**6.4**	**4.4**	**2.5**	**6.8**	**3.6**	**2.7**	**2.7**	**2.4**	**2.3**	**2.5**	**2.6**		**2.7**	**2.7**	2.7
Deficit in balance of goods and services	17.3	17.4	15.6			16.2	15.9	15.7	15.5	15.7	15.9		15.9	15.9	
Exports	47.4	44.8	42.2			43.7	43.7	44.2	44.4	44.3	44.1		44.1	44.1	
Imports	64.7	62.2	57.8			59.8	59.6	59.8	60.0	60.0	60.0		60.0	60.0	
Net current transfers (negative = inflow)	-18.5	-18.5	-18.7	-19.0	1.5	-19.0	-18.7	-18.7	-18.7	-18.7	-18.8		-18.7	-18.7	-18.7
of which: official	-0.6	-0.4	-0.2			-0.8	-0.8	-0.8	-0.8	-0.8	-0.8		-0.8	-0.8	
Other current account flows (negative = net inflow)	7.6	5.6	5.6			5.5	5.5	5.5	5.5	5.5	5.5		5.5	5.5	
Net FDI (negative = inflow)	**-6.7**	**-5.4**	**-3.7**	**-5.6**	**1.4**	**-4.9**	**-4.8**	**-4.9**	**-5.0**	**-5.1**	**-5.2**		**-5.2**	**-5.2**	-5.2
Endogenous debt dynamics 2/	**-1.7**	**-0.7**	**0.2**			**0.1**	**0.3**	**0.2**	**0.3**	**0.1**	**0.1**		**0.0**	**0.0**	
Contribution from nominal interest rate	0.6	1.1	1.3			1.4	1.6	1.6	1.7	1.6	1.5		1.2	0.7	
Contribution from real GDP growth	-1.0	-1.3	-1.3			-1.4	-1.3	-1.4	-1.4	-1.4	-1.4		-1.2	-0.6	
Contribution from price and exchange rate changes	-1.3	-0.6	0.2			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**2.8**	**1.0**	**0.5**			**4.5**	**1.5**	**2.8**	**2.6**	**1.8**	**1.7**		**1.2**	**1.5**	
of which: exceptional financing	-0.1	0.0	-0.1			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	31.8			34.6	34.1	34.2	34.3	33.5	32.6		26.6	16.1	
In percent of exports	...	...	75.3			79.2	78.1	77.5	77.2	75.6	73.8		60.4	36.6	
PV of PPG external debt	**...**	**...**	**25.3**			**28.6**	**28.5**	**28.9**	**29.0**	**28.2**	**27.3**		**21.6**	**11.3**	
In percent of exports	**...**	**...**	**59.9**			**65.5**	**65.2**	**65.4**	**65.3**	**63.7**	**62.0**		**48.9**	**25.6**	
In percent of government revenues	**...**	**...**	**79.5**			**91.4**	**89.7**	**91.1**	**90.9**	**88.6**	**85.5**		**67.7**	**35.4**	
Debt service-to-exports ratio (in percent)	**21.1**	**22.7**	**26.6**			**16.1**	**16.0**	**15.4**	**19.8**	**15.1**	**16.6**		**19.1**	**14.3**	
PPG debt service-to-exports ratio (in percent)	**1.9**	**3.6**	**4.2**			**4.8**	**5.3**	**5.3**	**10.4**	**5.8**	**7.4**		**10.2**	**5.9**	
PPG debt service-to-revenue ratio (in percent)	**3.0**	**5.3**	**5.5**			**6.7**	**7.3**	**7.4**	**14.4**	**8.0**	**10.2**		**14.2**	**8.2**	
Total gross financing need (Billions of U.S. dollars)	2.2	2.3	2.5			1.4	1.5	1.3	1.8	1.3	1.6		2.4	2.8	
Non-interest current account deficit that stabilizes debt ratio	5.6	5.1	3.0			0.3	3.1	1.9	2.1	3.1	3.4		4.0	3.6	
Key macroeconomic assumptions															
Real GDP growth (in percent)	3.1	3.6	3.6	3.3	2.2	4.0	3.6	3.8	3.8	3.8	3.8	3.8	3.8	3.4	3.8
GDP deflator in US dollar terms (change in percent)	3.6	1.6	-0.5	3.7	3.9	2.0	0.3	-1.0	-1.0	-0.5	0.2	0.0	1.5	3.1	1.5
Effective interest rate (percent) 5/	1.7	3.2	3.7	2.6	0.6	4.2	4.3	4.4	4.5	4.2	4.2	4.3	3.9	3.5	3.8
Growth of exports of G&S (US dollar terms, in percent)	5.8	-0.4	-2.9	4.9	13.4	9.8	4.0	3.9	3.4	3.0	3.7	4.6	5.3	6.6	5.3
Growth of imports of G&S (US dollar terms, in percent)	0.8	1.3	-4.2	5.2	16.1	9.9	3.5	3.2	3.1	3.4	4.1	4.5	5.3	6.6	5.3
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	0.3	13.7	18.8	2.6	20.3	14.7	11.7	-1.2	14.7	13.7
Government revenues (excluding grants, in percent of GDP)	30.3	30.3	31.8			31.3	31.8	31.7	31.9	31.9	32.0		31.9	31.9	31.9
Aid flows (in Billions of US dollars) 7/	0.3	0.3	0.3			0.3	0.2	0.2	0.2	0.2	0.2		0.3	0.5	
of which: Grants	0.2	0.2	0.2			0.1	0.2	0.2	0.2	0.2	0.2		0.2	0.4	
of which: Concessional loans	0.2	0.1	0.1			0.2	0.0	0.0	0.1	0.0	0.1		0.0	0.1	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			0.7	0.9	1.2	0.8	1.1	1.1		0.6	1.0	0.9
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			10.3	36.9	35.4	16.2	42.0	32.5		15.0	35.4	45.2
Memorandum items:															
Nominal GDP (Billions of US dollars)	19.7	20.7	21.4			22.7	23.6	24.2	24.9	25.7	26.8		34.4	58.4	
Nominal dollar GDP growth	6.8	5.3	3.1			6.1	3.9	2.8	2.8	3.3	4.1	3.8	5.3	6.6	5.3
PV of PPG external debt (in Billions of US dollars)			5.3			6.4	6.5	6.8	7.0	7.1	7.2		7.3	6.5	
(PVt-PVt-1)/GDPt-1 (in percent)						**5.4**	0.6	1.2	0.9	0.3	0.3	1.4	0.2	-0.2	-0.1
Gross workers' remittances (Billions of US dollars)	3.4	3.6	3.8			4.2	4.2	4.4	4.5	4.6	4.8		6.2	10.5	
PV of PPG external debt (in percent of GDP + remittances)	...	...	21.4			24.2	24.1	24.5	24.6	23.9	23.2		18.3	9.6	
PV of PPG external debt (in percent of exports + remittances)	...	...	42.0			46.1	46.2	46.5	46.5	45.3	44.0		34.7	18.2	
Debt service of PPG external debt (in percent of exports + remittanc	...	...	2.9			3.4	3.8	3.8	7.4	4.1	5.2		7.3	4.2	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table AII.2. Honduras: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2017-37

(In percent)

		Projections						
	2017	2018	2019	2020	2021	2022	**2027**	2037
PV of debt-to-GDP+remittances ratio								
Baseline	24	24	24	25	24	23	**18**	10
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	24	26	28	30	31	32	**37**	39
A2. New public sector loans on less favorable terms in 2017-2037 2/	24	24	25	26	26	25	**22**	16
A3. Customized 7/	24	24	25	26	26	25	22	10
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	24	24	25	25	24	24	**19**	10
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	24	27	35	36	35	34	**29**	13
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	24	24	24	24	23	22	**18**	9
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	24	25	27	27	26	25	**20**	10
B5. Combination of B1-B4 using one-half standard deviation shocks	24	25	29	29	28	28	**22**	11
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	24	32	32	32	31	30	**24**	12
PV of debt-to-exports+remittances ratio								
Baseline	46	46	46	46	45	44	**35**	18
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	46	49	53	57	60	62	**73**	80
A2. New public sector loans on less favorable terms in 2017-2037 2/	46	46	47	49	48	48	**43**	30
A3. Customized 7/	55	55	57	58	58	58	49	24
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	46	45	45	45	44	43	**34**	18
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	46	57	79	79	78	77	**64**	30
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	46	45	45	45	44	43	**34**	18
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	46	48	51	50	50	48	**39**	19
B5. Combination of B1-B4 using one-half standard deviation shocks	46	50	60	60	59	58	**47**	23
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	46	45	45	45	44	43	**34**	18
PV of debt-to-revenue ratio								
Baseline	91	90	91	91	89	86	**68**	35
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	91	95	102	108	112	115	**131**	137
A2. New public sector loans on less favorable terms in 2017-2037 2/	91	89	93	95	95	94	**83**	59
A3. Customized 7/	90	76	79	80	81	80	68	33
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	91	90	93	93	91	88	**69**	36
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	91	102	131	131	130	127	**106**	49
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	91	88	88	88	86	83	**66**	34
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	91	92	99	99	97	94	**76**	38
B5. Combination of B1-B4 using one-half standard deviation shocks	91	94	106	106	104	101	**82**	40
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	91	125	127	126	123	119	**94**	49

Table AII.2. Honduras: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2017-37 (concluded)

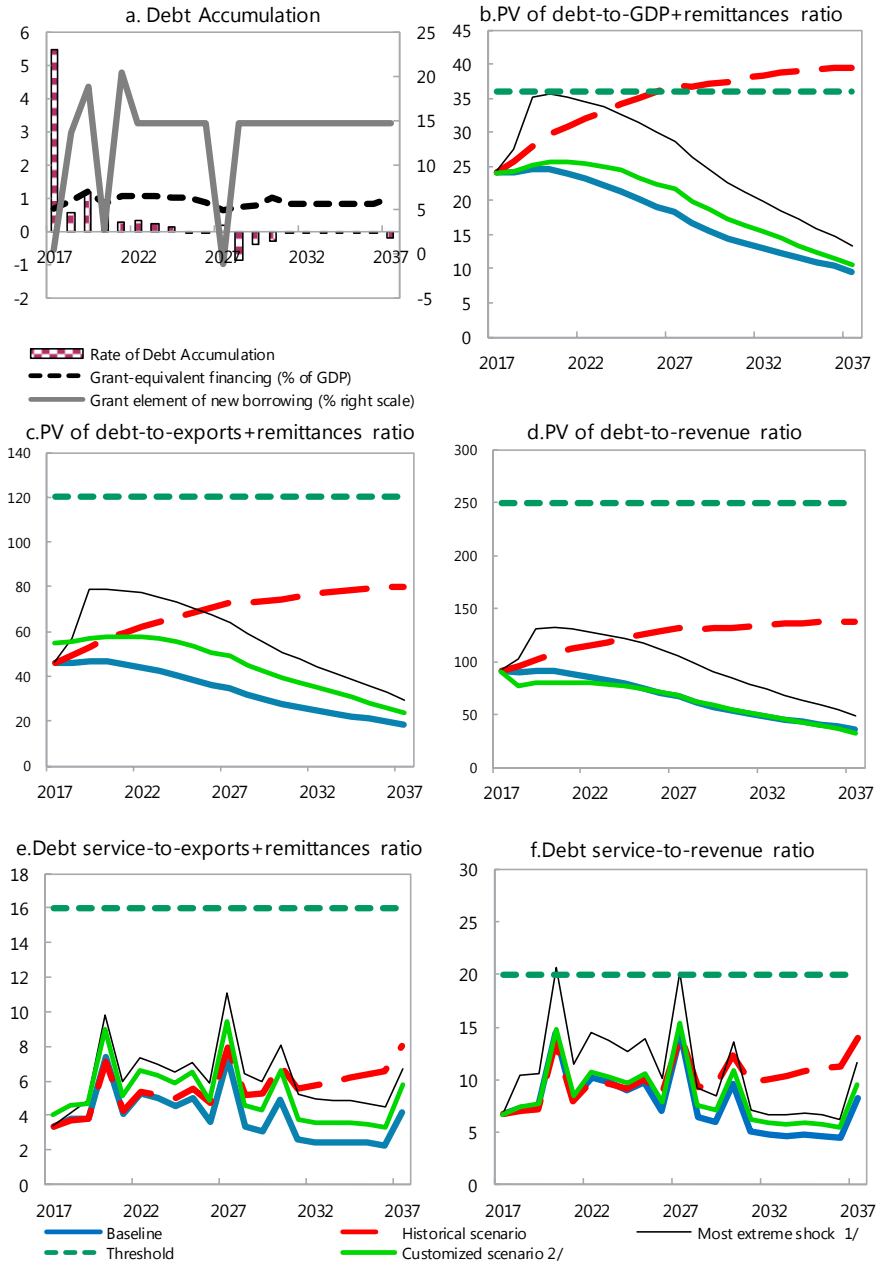
(In percent)

	Projections							
	2017	2018	2019	2020	2021	2022	2027	2037
Debt service-to-exports+remittances ratio								
Baseline	3	4	4	7	4	5	**7**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	3	4	4	7	4	5	**8**	8
A2. New public sector loans on less favorable terms in 2017-2037 2/	3	4	3	7	3	4	**3**	3
A3. Customized 7/	4	5	5	9	5	7	9	6
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	3	4	4	7	4	5	**7**	4
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	3	4	5	10	6	7	**11**	7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	3	4	4	7	4	5	**7**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	3	4	4	8	4	5	**8**	5
B5. Combination of B1-B4 using one-half standard deviation shocks	3	4	4	8	5	6	**9**	5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	3	4	4	7	4	5	**7**	4
Debt service-to-revenue ratio								
Baseline	7	7	7	14	8	10	**14**	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	7	7	7	13	8	10	**14**	14
A2. New public sector loans on less favorable terms in 2017-2037 2/	7	7	6	13	7	8	**6**	5
A3. Customized 7/	7	7	8	15	8	11	15	9
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	7	8	8	15	8	11	**15**	9
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	7	7	8	16	10	12	**18**	11
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	7	7	7	14	8	10	**14**	8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	7	7	8	15	8	11	**15**	9
B5. Combination of B1-B4 using one-half standard deviation shocks	7	7	8	15	9	11	**16**	9
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	7	10	11	21	11	15	**20**	12
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	7	7	7	7	7	7	**7**	7

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.
7/ In the customized scenario, (i) real GDP growth is 2.6 percent in 2016 and 2.5 in 2017, and rises on average 0.25 percentage points per year to reach 3.5 percent in 2021, staying at this level for the remainder of the DSA period; (ii) financing conditions from external capital markets deteriorate permanently, raising the average interest rate on new external debt by 100 basis points; and (iii) the overall deficit of the CPS increases by 2 percent of GDP in 2015 compared to the baseline and then fades out partially at a constant rate of 15 percent during the DSA period.

Figure AII.1. Honduras: Indicators of Public and Publicly Guaranteed External Debt Including Remittances, 2017-37 1/

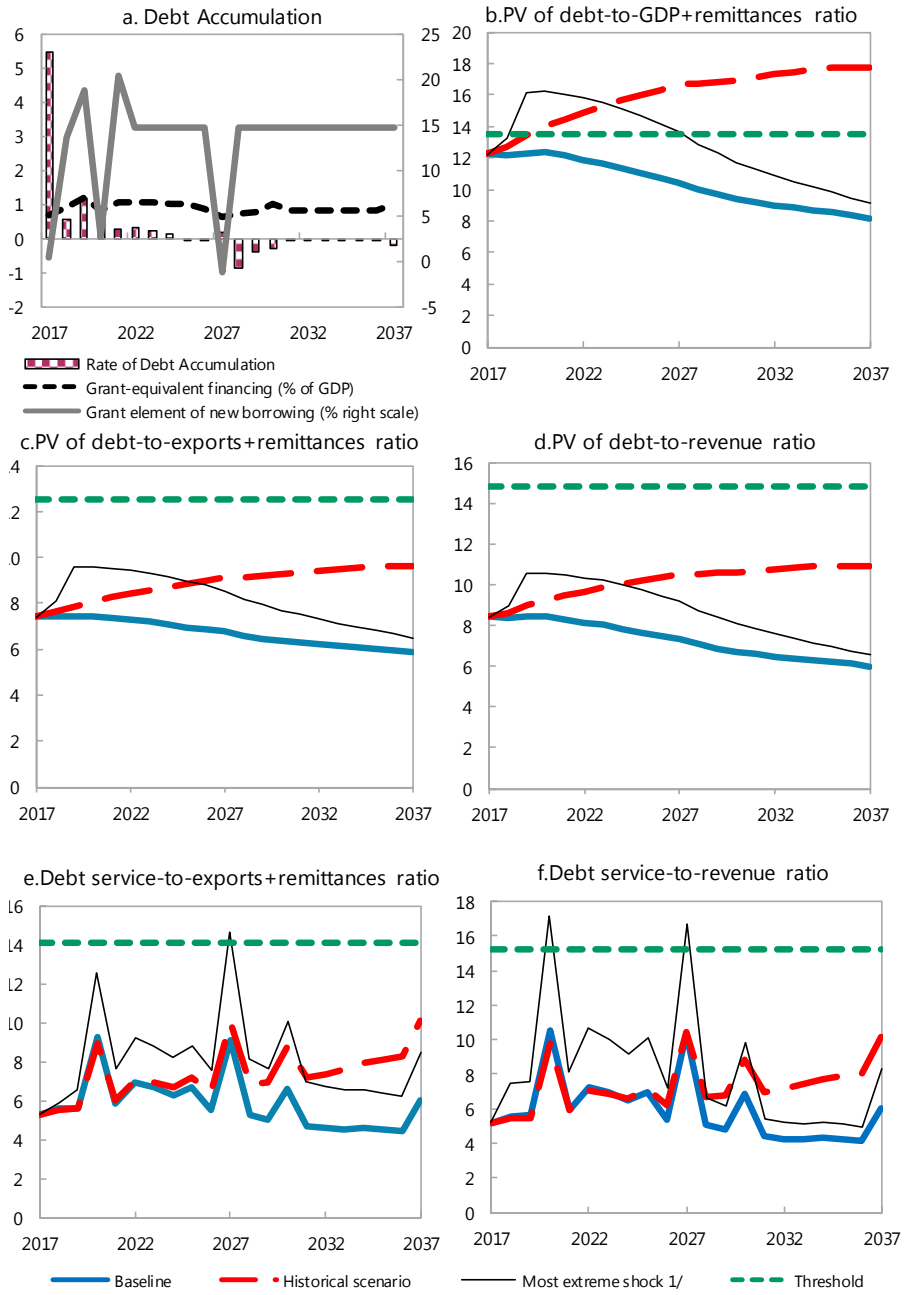


Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2027. In figure b. it corresponds to a Exports shock; in c. to a Exports shock; in d. to a Exports shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

2/ In the customized scenario, (i) real GDP growth is 2.6 percent in 2016 and 2.5 in 2017, and rises on average 0.25 percentage points per year to reach 3.5 percent in 2021, staying at this level for the remainder of the DSA period; (ii) financing conditions from external capital markets deteriorate permanently, raising the average interest rate on new external debt by 100 basis points; and (iii) the overall deficit of the CPS increases by 2 percent of GDP in 2015 compared to the baseline and then fades out partially at a constant rate of 15 percent during the DSA period.

Figure AII.2. Honduras: Probability of Debt Distress of Public and Publicly Guaranteed External Debt under Alternative Scenarios, Including Remittances, 2017-37 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2027. In figure b. it corresponds to a Exports shock; in c. to a Exports shock; in d. to a Exports shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Table AII.3. Honduras: Public Sector Debt Sustainability Framework, Baseline Scenario, 2014-37

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate					Projections				
	2014	2015	2016	Average 5/	Standard Deviation 5/	2017	2018	2019	2020	2021	2022	2017-22 Average	2027	2037	2023-37 Average
Public sector debt 1/	42.3	42.3	43.5			43.5	44.3	45.0	42.8	40.5	39.8		33.3	13.4	
of which: foreign-currency denominated	28.2	28.6	29.2			32.2	32.3	33.1	33.3	32.7	32.0		25.9	14.3	
Change in public sector debt	1.8	0.0	1.2			0.0	0.8	0.7	-2.2	-2.3	-0.7		-1.6	-2.4	
Identified debt-creating flows	3.3	0.2	1.8			-0.5	1.4	0.8	0.7	0.4	0.3		-0.4	-1.2	
Primary deficit 2/	3.6	0.3	0.3	3.2	2.1	0.1	0.2	-0.4	-0.3	-0.2	-0.2	-0.1	-0.4	-0.7	-0.5
Revenue and grants	29.4	29.9	30.7			30.3	30.7	30.9	31.1	31.2	31.3		31.0	31.0	
of which: grants	0.8	0.9	0.8			0.7	0.6	0.7	0.7	0.7	0.7		0.7	0.7	
Primary (noninterest) expenditure	33.0	30.3	31.0			30.4	30.8	30.5	30.8	30.9	31.2		30.6	30.3	
Automatic debt dynamics	-1.0	-0.6	1.2			-0.7	1.3	1.2	1.0	0.6	0.4		0.0	-0.5	
Contribution from interest rate/growth differential	-0.9	-0.2	0.4			-0.2	0.2	0.2	0.0	0.0	-0.1		-0.1	-0.3	
of which: contribution from average real interest rate	0.3	1.3	1.9			1.4	1.7	1.8	1.7	1.5	1.4		1.1	0.2	
of which: contribution from real GDP growth	-1.2	-1.5	-1.5			-1.7	-1.5	-1.6	-1.6	-1.6	-1.5		-1.3	-0.5	
Contribution from real exchange rate depreciation	-0.1	-0.4	0.8			-0.4	1.1	1.0	1.0	0.7	0.5		...	...	
Other identified debt-creating flows	0.7	0.4	0.2			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.7	0.4	0.2			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	-1.6	-0.2	-0.5			0.5	-0.7	-0.1	-2.8	-2.7	-1.0		-1.2	-1.2	
Other Sustainability Indicators															
PV of public sector debt	...	...	39.4			39.9	40.5	40.8	38.5	36.0	35.1		28.9	10.3	
of which: foreign-currency denominated	...	...	25.1			28.6	28.5	28.9	29.0	28.2	27.3		21.6	11.3	
of which: external	...	...	25.3			28.6	28.5	28.9	29.0	28.2	27.3		21.6	11.3	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 3/	3.4	4.6	3.2			7.4	3.7	4.3	6.6	4.4	5.1		5.7	0.2	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	128.6			131.7	132.0	132.0	123.7	115.5	112.1		93.1	33.4	
PV of public sector debt-to-revenue ratio (in percent)	...	...	131.9			134.6	134.8	135.0	126.4	118.1	114.6		95.2	34.1	
of which: external 4/	...	...	84.6			96.6	94.8	95.6	95.2	92.5	89.1		71.0	37.2	
Debt service-to-revenue and grants ratio (in percent) 5/	7.6	17.8	14.5			27.9	16.4	19.1	25.6	18.4	20.1		23.3	7.4	
Debt service-to-revenue ratio (in percent) 6/	7.8	18.3	14.9			28.5	16.8	19.5	26.2	18.8	20.5		23.8	7.5	
Primary deficit that stabilizes the debt-to-GDP ratio	1.9	0.4	-0.9			0.2	-0.6	-1.1	1.8	2.1	0.5		1.2	1.7	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	3.1	3.6	3.6	3.3	2.2	4.0	3.6	3.8	3.8	3.8	3.8	3.8	3.8	3.4	3.8
Average nominal interest rate on forex debt (in percent)	1.7	3.6	3.4	2.3	0.7	4.3	4.4	4.4	4.6	4.2	4.2	4.4	3.9	3.4	3.7
Average nominal interest rate on domestic debt (in percent)	9.7	11.0	14.2	10.8	2.9	12.6	14.1	14.2	12.0	13.8	14.6	13.5	13.6	11.6	13.1
Average real interest rate (in percent)	0.8	3.1	4.7	2.1	1.9	3.4	4.1	4.3	3.9	3.8	3.6	3.9	3.4	1.3	3.1
Average real interest rate on foreign-currency debt (in percent)	-1.7	-1.0	-1.2	-1.6	0.5	-2.1	-2.1	-2.1	-1.9	-1.9	-2.0	-2.0	-2.0	-2.0	-2.0
Average real interest rate on domestic debt (in percent)	2.7	4.2	10.1	5.0	4.1	6.3	9.9	9.8	7.7	9.4	10.2	8.9	9.3	2.1	8.3
Real exchange rate depreciation (in percent, + indicates depreciation	-0.5	-1.3	2.8	-1.6	3.7	-1.5	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	6.8	6.4	3.7	5.6	2.2	5.9	3.8	4.0	4.0	4.0	4.0	4.3	4.0	9.3	4.4
Growth of real primary spending (deflated by GDP deflator, in percer	-5.5	-5.0	6.1	-0.4	3.2	2.0	5.0	2.7	4.8	4.3	4.7	3.9	3.5	3.8	3.6
Grant element of new external borrowing (in percent)	...	...	...	...	...	0.3	13.7	18.8	2.6	20.3	14.7	11.7	-1.2	14.7	...

Sources: Country authorities; and staff estimates and projections.
1/ Gross debt of the Combined Public Sector
2/ As defined in the fiscal accounts
3/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
4/ Revenues excluding grants.
5/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table AII.4. Honduras: Sensitivity Analysis for Key Indicators of Public Debt 2017-37

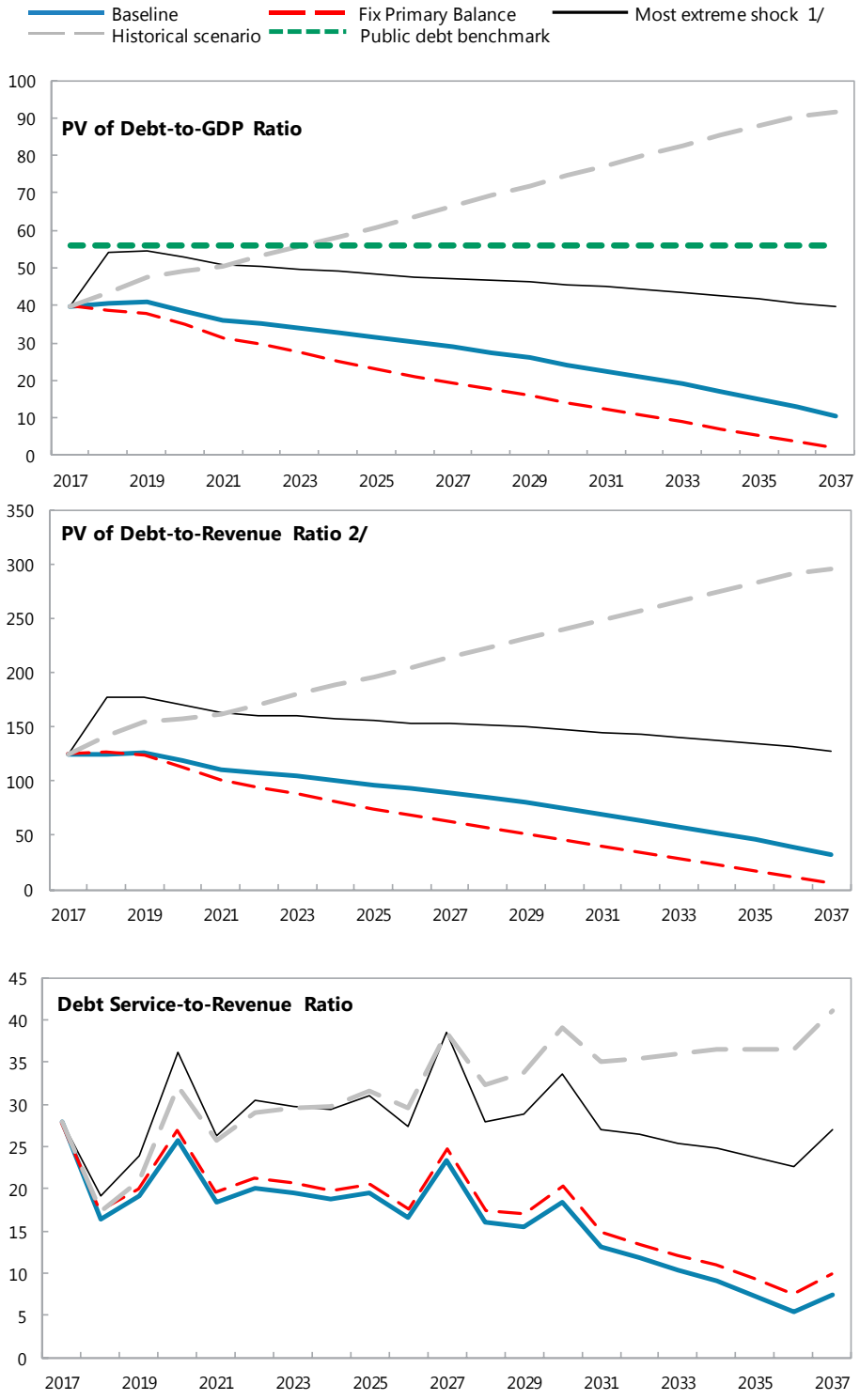
	Projections							
	2017	2018	2019	2020	2021	2022	2027	2037
PV of Debt-to-GDP Ratio								
Baseline	40	40	41	38	36	35	29	10
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	40	43	47	49	50	53	66	92
A2. Primary balance is unchanged from 2017	40	39	38	35	31	29	19	2
A3. Permanently lower GDP growth 1/	40	41	42	40	39	39	41	53
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	40	42	45	44	44	45	48	50
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	40	45	51	49	47	47	42	31
B3. Combination of B1-B2 using one half standard deviation shocks	40	45	50	49	48	48	48	45
B4. One-time 30 percent real depreciation in 2018	40	54	55	53	51	50	47	40
B5. 10 percent of GDP increase in other debt-creating flows in 2018	40	50	50	48	46	46	42	30
PV of Debt-to-Revenue Ratio 2/								
Baseline	125	125	126	118	111	108	89	32
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	125	142	154	158	162	170	214	295
A2. Primary balance is unchanged from 2017	125	126	123	112	100	94	63	6
A3. Permanently lower GDP growth 1/	125	133	134	129	124	124	133	170
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	125	137	146	143	140	142	154	162
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	125	148	166	158	151	149	137	100
B3. Combination of B1-B2 using one half standard deviation shocks	125	146	163	158	154	154	155	143
B4. One-time 30 percent real depreciation in 2018	125	177	177	169	163	160	152	128
B5. 10 percent of GDP increase in other debt-creating flows in 2018	125	163	163	155	148	146	134	98
Debt Service-to-Revenue Ratio 2/								
Baseline	28	16	19	26	18	20	23	7
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	28	17	21	32	26	29	38	41
A2. Primary balance is unchanged from 2017	28	17	20	27	20	21	25	10
A3. Permanently lower GDP growth 1/	28	17	21	29	22	24	32	27
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-2019	28	18	21	31	24	27	34	27
B2. Primary balance is at historical average minus one standard deviations in 2018-2019	28	17	21	34	27	26	31	21
B3. Combination of B1-B2 using one half standard deviation shocks	28	18	22	34	27	27	33	25
B4. One-time 30 percent real depreciation in 2018	28	19	24	36	26	30	38	27
B5. 10 percent of GDP increase in other debt-creating flows in 2018	28	17	22	37	23	27	31	20

Sources: Country authorities; and staff estimates and projections.

1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.

2/ Revenues are defined inclusive of grants.

Figure AII.3. Honduras: Indicators of Public Debt Under Alternative Scenarios, 2017-37 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2027.
2/ Revenues are defined inclusive of grants.